BOARD OF SECONDARY EDUCATION, KARACHI

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2025

CHEMISTRY

Past Paper 2025

Retyped faithfully from the original board paper — same questions, no changes.

SECTION "A" (COMPULSORY) MULTIPLE CHOICE QUESTIONS (MCQs)

Maximum Marks: 12

Note: Note: (i) Attempt all the parts of this section. (ii) Do not copy down the part of question. Write only the answer against the proper number of the part according to the question paper. (iii) Each part carries 1 mark.

Q1. 1. Choose the correct answer for each from the given options:

Q1. The number of moles equivalent to 8g of CO2 are: : 8 i) CO2)

(A) 0.15
(B) 0.18
(C) 0.18
(D) 0.15

Q2. Molecular mass of Nitric Acid (HNO3) is: : (HNO3) (ii)

(A) 60 a.m.u.
(B) 50 a.m.u.
(C) . 50a.m.u
(D) . 60a.m.u

Q3. The branch of Chemistry in which purple iodine vapours (iii) appear when solid iodine is heated, is called:

(A) Environmental Chemistry
(B) Biochemistry
(C) Medicinal Chemistry
(D) Inorganic Chemistry

Q4. This isotope is used to stimulate heart beat in heart pace pace maker (iv) maker:

(A) Plutonium – 238
(B) Carbon – 14 :
(C) Cobalt – 60
(D) Phosphorous – 32 14 –

Q5. All transition elements are: : (v)

(A) Gases
(B) Metals
(C) Non-metals
(D) Metalloids

Q6. When an electronegative element combines with an (vi) electropositive element, the type of bond form:

(A) Covalent bond
(B) Ionic bond
(C) Polar bond
(D) Coordinate covalent bond

Q7. The compound not soluble in water: : (vii)

(A) NaCl
(B) C6H6
(C) C6H6
(D) NaCl

Q8. One atmospheric pressure is equal to: : (viii)

(A) 10325 Pascal
(B) 106075 Pascal
(C) 106075
(D) 10325

Q9. Photosynthesis is a/an: : (ix)

(A) Oxidation Reaction
(B) Redox Reaction
(C) Reduction Reaction
(D) Neutralization Reaction

Q10. Hardness of metal is measured by: : (x)

(A) pH scale
(B) likert scale
(C) pH
(D) rector scale

Q11. Fluorine exists in the form of:

(A) Solid
(B) Liquid
(C) Gas
(D) All of these

Q12. KCl)

(A) unsaturated solution
(B) super saturated solution
(C) diluted solution
(D) suspension

SECTION “B” : (SHORT-ANSWER QUESTIONS) (3) (8) :

Maximum Marks: 24

Note: Note: Attempt any EIGHT(8) questions from this section. Each question carries three (3) marks.

Q2. Write down about the applications of Chemistry in daily life.

Q3. Differentiate between Molecular and Empirical formula.

Q4. Complete and balance the following equations:

(i) KNO3 → → i) KNO3)

(ii) Ca + H2O →

(iii) $NaHCO_3 \rightarrow \rightarrow$ ii) $Ca + H_2O$)

Q5. Write down the electronic configuration of Na, Mg and Al While Na=11 Al Na Mg their atomic number are Na=11, Mg = 12 and Al=13. Al=13 Mg=12

Q6. The pressure of a sample gas is 8 atm and the volume is 15 liters. If 6 atm 15 8 atm the pressure is reduced to 6 atm, what will be its new volume?

Q7. What is meant by an electronegativity? Describe its trend in group and period with examples.

Q8. What is meant by lone pair and unpaired electrons? How many lone pair and unpaired electrons are there in Nitrogen atom?

Q9. Define the following terms: (iii) (ii) (i) :

(i) Free radical

(ii) Solubility

(iii) Alloy

Q10. Calculate the percentage concentration (m/m) of the solution by 110 15 dissolving 15g salt in 110g water. (m/m)

Q11. Write down the chemical formulae and one use of each of the : following compounds: (iii) (ii) (i)

(i) Soda Ash

(ii) Baking Soda

(iii) Washing Soda

Q12. What is redox reaction? Give an example of it with equation and also identify the oxidizing and reducing agents in the equation.

Q13. Give reasons:

(i) How does electron differ from neutron? (i)

(ii) Why the rate of diffusion of gases is rapid than that of liquids? (ii)

SECTION “C” : (DESCRIPTIVE-ANSWER QUESTIONS) (6) (4) :

Maximum Marks: 24

> Note: Note: Attempt any FOUR (4) questions from this Section. Each question carries Six (6) marks.

Q14. How did Rutherford present the atomic model? Describe his experiment in detail and explain the atomic model presented by him.

Q15. Why do some molecules form covalent bond? Write its three types with examples.

Q16. Describe the construction of a dry cell with diagram. Explain its working with the help of chemical equations.

Q17. Compare properties of Solution, Suspension and Collides. (Write eight points of each)

Q18. How elements are arranged in periodic table? Describe groups and periods in periodic table.

Q19. What is Allotropy? Describe different allotropic forms of Carbon.